

# Training Locomotor Control Policies on Pathological Gait Kinematics Reveals Emergent Clinical Signatures and Limits of Motor-Level Mechanism Injection

Wafi Sharif<sup>1\*</sup>, Miles Smith<sup>2</sup>

<sup>1</sup>Queens High School for the Sciences, Jamaica, NY

<sup>2</sup>Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, MA

\*Corresponding Author: Wafi Sharif, wafisharif0@gmail.com

**Abstract**--Stroke is the leading cause of gait impairment in adults, and reinforcement-learning simulations are increasingly being used to study pathological locomotion. But these simulations are rarely checked against real patient movement, so a simulated walk that looks reasonable may not actually behave like a real patient. This paper builds locomotor controllers and validates them against gait data across five clinical populations: stroke, amyotrophic lateral sclerosis (ALS), Huntington's disease (HD), Parkinson's disease (PD), and healthy controls. For stroke, where detailed motion and muscle-activity data exist, we trained a controller directly on that data. It was balanced against perturbations much more reliably than a comparison controller built only to impose a stroke-like weakness pattern, indicating that the advantage comes from training on real motion. It also yielded a documented post-stroke walking pattern, stiff-knee gait, that it was never trained to produce. By contrast, ALS, HD, and PD datasets only capture step timing, so we built controllers by adding calibrated impairments such as weakness or motor noise onto a baseline, then analyzed how close their timing patterns landed to each real population. This moved the simulated patterns closer to real ones for many features, but not all. Collectively, these results show that training on real movement data can build a pathological controller that can effectively resemble its corresponding pathology, while approximating a pathology with hand-picked impairments only works partially.

**Keywords:** musculoskeletal simulation, reinforcement learning, gait analysis, biomechanics, principal component analysis

## I. INTRODUCTION

Stroke is a leading cause of long-term motor disability, with approximately 104.8 million people living with its effects and 13.2 million new cases yearly [1]. Unilateral gait impairment is a common consequence, with an estimated 46% of first-time survivors unable to walk after stroke onset [2].

With such drastic effects, reinforcement-learning policies are increasingly being used to represent pathological motion, like that of stroke, in hopes of informing rehabilitation strategies and assistive-device control [3], [4]. However, control policies

used for understanding pathological movement rarely are trained on data from pathological populations [5, 6, 7]. In many cases, significant kinematic and electrophysiological data have already been collected but are not used in any way to inform policy design, suggesting a gap where pathological movement data can be used to train control policies for musculoskeletal models

Here, we investigate the effect of different training datasets on the optimized motor control policy using a musculoskeletal model actuated with 80 lower-body muscles. Specifically, we use post-stroke gait data to train a policy, then compare it against existing policies that use synthetic deficits to represent pathologies. We explore if training with kinematic imitation yields clinical mechanisms, and whether pathological controllers which, by contrast, only inject synthetic impairment mechanisms behave similarly to real patients.

This paper shows that a locomotor control policy built and validated on clinical data outperforms those built on synthetic deficits and reproduces clinical mechanisms. We still show that the construction can be extended to pathologies with sparser data, but only up to a certain point; synthetic impairment injection can capture some clinical patterns but fails to reach many others, which reveals the limitations of this approach. Adding a tuned impairment onto a healthy policy partially fixes the mismatch between simulated and real gait timing, but not all of it. No matter how the training is adjusted, many features never match real patients, showing that this approach has a hard limit. This paper contributes a framework for pathologically-informed locomotor policies, and a thorough account of where motor-level mechanism injection reaches its limits.

## II. METHODS

### *Musculoskeletal model and simulation environment*

The policies constructed were tested on the MuJoCo/MyoSuite myoLegWalk-v0 environment [8], an 80-muscle, three-dimensional lower-limb model.

Control policies were constructed using gait data from publicly accessible datasets. A full-body motion-capture dataset from Van Criel et al. [9] covers 138 able-bodied adults and 50 stroke survivors, with 3D joint kinematics and electromyography (EMG) data from seven leg muscles,

supplying the joint-tracking policy’s target and its validation curves. PhysioNet’s Gait Dynamics in Neurodegenerative Disease database (GaitNDD) [10], [11] provides per-stride force-sensor timing across 16 healthy, 15 Parkinson’s disease (PD), 20 Huntington’s disease (HD), and 13 amyotrophic lateral sclerosis (ALS) subjects, including stride, swing, stance, and double-support duration per step, but no joint angles or EMG.

#### ***Using reinforcement learning to train model control policies***

Each policy used the same model, a randomized start pose, and a 1000-step episode limit, and was trained and evaluated across 20 seeds. Our random-activation baseline is an untrained floor policy. Meanwhile, our other three policies are trained with Maximum a Posteriori Policy Optimization (MPO) [12], [13] via the Tonic RL library with Differential Extrinsic Plasticity (DEP)-based exploration. It uses a two-hidden-layer, 1024-unit actor-critic network (actor/critic learning rates  $3.53 \times 10^{-5} / 6.08 \times 10^{-5}$ , replay buffer  $1 \times 10^6$ , batch size 256), differing only in reward. The synthetic-reward policy uses DEP-RL’s out-of-the-box synthetic reward, with no reference to human movement (its motion looks realistic but is technically not representative of a human’s). The literature-reward policy trains on Schumacher et al.’s literature-exact reward [8]; it has a velocity term with effort and pain costs, and they report a 43% ( $\pm 5\%$ ) kinematic match on Hyfydy (a separate musculoskeletal physics engine). The joint-tracking policy adds paretic and non-paretic hip/knee/ankle/pelvis targets, each of which is weighted based on how well it correlates with real patient data, to the literature-reward’s terms.

#### ***Calibrated mechanism injection for sparse-data pathologies***

GaitNDD’s timing-only data rules out kinematic imitation for these three. So, with an extension of the stroke construction to sparser data, each impairment controller is built by injecting a synthetic mechanism onto the literature-reward substrate. Namely, we impose strength scaling to build a weakness controller (calibrated against ALS gait timing), additive action-space (motor) noise for a motor-noise controller (calibrated against HD gait timing), and additive observation-space (feedback) noise for a feedback-noise controller (calibrated against PD gait timing). Each mechanism’s free parameter comes from a predefined coordinate-search grid, maximizing cosine similarity between the controller’s direction-of-change vector (stride duration, double-support %, left/right stance %) and the same vector from real per-subject GaitNDD data. A healthy variant (uniform strength  $\times 0.85$ ) anchors all five conditions to one substrate.

#### ***Reachability screen and training-based interventions***

Before fitting mechanism parameters harder, we explored whether the substrate could reach real targets at all. A grid sweep extended each mechanism’s parameter past its calibration range (8 seeds/point, reset\_type = ‘random’) to track each feature’s stable-operating envelope. A 40-point factorial screen (strength  $\times$  noise  $\times$  jitter, 6 seeds each) assessed the effect of combined mechanisms. Interventions were run against the target from the literature-reward substrate: a windowed squared-error reward-shaping fine-tune (200,000 steps); a five-variant sweep of that term’s weight, window, and formulation, as done in reward-transformation literature [14]; and an adversarial motion-prior fine-tune [15], with its discriminator stabilized

with an R1 gradient penalty, which is a regularization technique that stabilizes adversarial/GAN-style training.

#### ***Outcome metrics***

Joint-angle range, ground-reaction-force shape during stance, and stance/swing ratio were assessed. Ground contact is read from foot-sensor force above 1 N. Stroke asymmetry uses the raw symmetry index of Robinson et al. [16]. Knee flexion during swing is measured as swing-isolated peak values; these are aggregated per episode, so non-independent swing phases are not treated as separate samples, but instead as full gait-cycle curves reconstructed from heel strikes.

#### ***Statistical methods***

Principal components were fitted onto the GaitNDD data, with four timing features across all four non-stroke conditions. The joint-tracking policy is validated separately, against the gait data from stroke survivors. We also control offset between the substrate and real data by only looking at the direction each simulated disease moves away from simulated-healthy, compared to the direction the matching disease moves away from real-healthy. We use a binomial test to determine whether the total number of matches beats chance; and McNemar’s exact test to determine whether calibration improves match count. Group comparisons for step-survival, joint-kinematic, and muscle-activation measures use two-sided Mann-Whitney U tests, since these measures are continuous and not normally distributed [17]; meanwhile, stance-pattern proportions use Fisher’s exact test, since these are binary hit/miss outcomes [18]. Rate estimates (fall rate, perturbation-recovery rate) carry 10,000-resample bootstrap 95% confidence intervals; the perturbation sweep’s original 8 trials/cell were supplemented with 16 more at the four magnitude cells that set the headline “maximum tolerated push” figure, before intervals were finalized.

#### ***Protocol for testing control policies***

The flat-ground condition, which uses flat terrain and no external perturbations, reports fall rate, distance, and gait-timing signature; this is averaged over 20 simulated walking trials per controller. Each controller is then tested using a perturbation protocol. Specifically, perturbations are introduced as a 0.1 s force applied at the pelvis ranging from 200 to 800 N (Fig. 1). Each perturbation lands at the instant of single- or double-support onset, as prior literature proves that the support phase that a perturbation occurs in affects its severity [10]. We evaluate each controller’s recovery based on peak torso deviation, and terrain transfer uses MyoSuite’s native rough, hilly, and stairs generators [8].

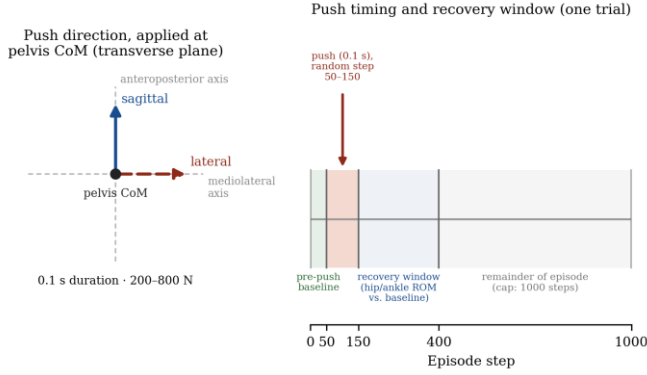


Fig. 1. Perturbation protocol: push direction and magnitude applied at the pelvis center of mass (left), and push timing relative to the pre-push baseline, recovery window, and full-episode cap (right).

### III. RESULTS

#### Reward design tradeoffs

Figure 2(a) summarizes this comparison. From no human-movement reference to a literature-based velocity/effort/pain reward [13], to joint-tracking terms on top of that, we use this progression to test if rewarding human-likeness sacrifices robustness. The synthetic-reward and literature-reward policies were both successful in number of steps completed (734.5 vs. 840.6, respectively;  $p = 0.99$ ). The literature-reward policy has the best human resemblance; its right foot shows the double-hump ground-reaction-force pattern in 44% of stances, compared to the synthetic-reward's 16.2% and joint-tracking policy's 1.4% ( $p < 0.0001$  both). Furthermore, its stance/swing ratio (49.6%) is closest to the real target of approximately 60/40 ( $p < 0.0001$  both). The joint-tracking policy survives every step of every trial (1000/1000, SD = 0; 12.31 m at 1.262 m/s against a 1.2 m/s target).

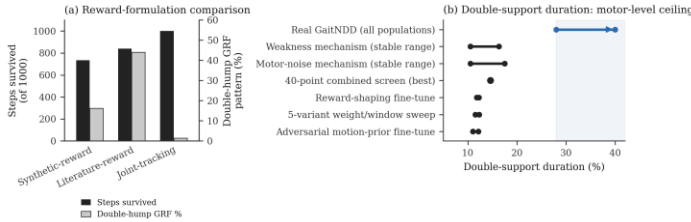


Fig. 2. (a) Steps survived and double-hump GRF% across the three reward formulations. (b) Double-support duration reached by each mechanism and fine-tuning intervention, against the real GaitNDD range (blue).

#### Paretic-knee asymmetry and an emergent clinical stiff-knee gait mechanism

We then check if the joint-tracking policy's paretic/non-paretic asymmetry matches the real asymmetry pattern. Simulated paretic/non-paretic asymmetry matches the target's direction and magnitude for hip ( $r = 0.740, 0.877$ ) and ankle ( $r = 0.487, 0.688$ ), but not the knee ( $r = 0.082, -0.140$ ): the paretic knee stays near full extension for nearly the whole episode. Swing-isolated peak flexion shows the same asymmetry in a single number (Table I): the synthetic-reward policy's paretic/non-paretic split is modest ( $74.4^\circ$  vs.  $64.1^\circ$ ), while the joint-tracking policy's split is more extreme, with its paretic-side mean ( $0.5^\circ$ ) sitting well below its non-paretic mean ( $79.7^\circ$ ). Figure 2 shows

this across the full gait cycle, as the real paretic curve flexes normally through swing toward a  $\sim 130^\circ$  peak, while the joint-tracking policy's paretic curve stays flat near  $0^\circ$  throughout, even though its target specified paretic-side flexion ( $49\text{--}132^\circ$ ).

Figure 3(b) summarizes the model's simulated muscle activation. Paretic-side knee extensors are more active during swing (rectus femoris 0.845 vs. 0.597; vastus lateralis 0.657 vs. 0.371) and flexors are suppressed (biceps femoris 0.164 vs. 0.416; semitendinosus 0.134 vs. 0.462). This is similar to the extensor-overactivity, flexor-suppression pattern that is attributed to reduced swing-phase knee flexion in the stiff-knee gait mechanism in clinical literature [14]. Therefore, we show that a reinforcement-learning policy with no reflex circuitry and activation-timing reward arrived at the muscle-recruitment strategy that produces that outcome, across measures excluded from training.

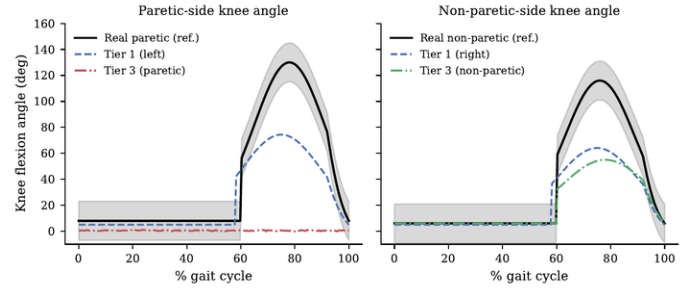


Fig. 3. (a) Phase-aligned knee-angle curves across a full gait cycle: real paretic/non-paretic reference (black,  $\pm 1$  SD) vs. synthetic-reward and joint-tracking policy.

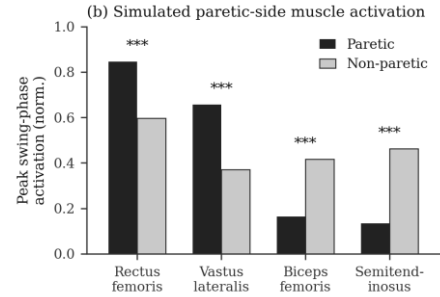


Fig. 3. (b) Simulated paretic-side muscle activation during swing, paretic vs. non-paretic (mean, 20 seeds; \*\*\*  $p < 0.001$ , all four muscles).

#### Double-support duration is limited by motor-level injection

Across the stable range of the ALS-strength and HD-noise mechanisms, simulated double-support duration never exceeds  $\sim 17\%$  (10.5–16.3% and 10.5–17.5%), while every real GaitNDD population sits at 28% or higher. The 40-point combined-mechanism screen found that the best stable combination reaches 14.57% double support. We tried three more ways to close this gap. Specifically, a 200,000-step tweak that penalized any miss of target duration (12.27% to 11.80%); five variations of that penalty's strength/timing (11.5–12.3%); and a method rewarding realistic motion (11.05–12.13%, up to 80,000 steps).

#### Cross-population robustness

Table II summarizes the common-environment comparison. At the decisive 400 N sagittal, single-support-onset cell, the joint-tracking policy’s recovery rate (0.958, CI [0.875, 1.0],  $n = 24$ ) clears every other controller’s upper bound (Healthy 0.55 [0.32, 0.73]; motor-noise 0.50 [0.29, 0.71]; feedback-noise 0.25 [0.08, 0.42]; weakness 0.10 [0.00, 0.25]). Terrain transfer fails almost universally off flat ground (0% on hilly and stairs for all five; only Healthy clears rough terrain, at 5%).

To separate a disease effect from a construction-method effect, we built a matched synthetic-stroke controller on the literature-

reward substrate: unilateral (paretic-leg) strength scaling, calibrated by maximizing cosine similarity to Van Crielinge ankle/knee/hip ROM-asymmetry. When tested, its recovery rate at 400 N sagittal, single-support (0.50 [0.29, 0.71]) collapses into Healthy/motor-noise range. Its asymmetry direction differs from the joint-tracking policy; cosine similarity to the real three-joint vector is +0.40 [+0.37, +0.42] for the injected controller, versus -0.69 [-0.70, -0.68] for joint-tracking.

Table I. Swing-phase peak knee flexion ( $^{\circ}$ ), paretic vs. non-paretic side.

| Policy           | Paretic ( $^{\circ}$ ) | Non-paretic ( $^{\circ}$ ) |
|------------------|------------------------|----------------------------|
| Synthetic-reward | 74.4                   | 64.1                       |
| Joint-tracking   | 0.5                    | 79.7                       |

Table II. Common-environment comparison across five controllers (20 seeds/condition; 95% confidence interval in brackets).

| Controller     | Fall rate, flat   | Distance (m) | Double-support % | Recovery @ 400N sag.-single |
|----------------|-------------------|--------------|------------------|-----------------------------|
| Healthy        | 0.05 [0.00, 0.15] | 13.43        | 13.07            | 0.55 [0.32, 0.73]           |
| Weakness       | 0.95 [0.85, 1.00] | 5.64         | 13.43            | 0.10 [0.00, 0.25]           |
| Motor noise    | 0.05              | 13.55        | 13.75            | 0.50 [0.29, 0.71]           |
| Feedback noise | 0.80 [0.60, 0.95] | 6.88         | 13.57            | 0.25 [0.08, 0.42]           |
| Stroke         | 0.00              | 12.77        | 3.63             | 0.958 [0.875, 1.00]         |

### Cross-population PCA validation

Fig. 4 projects every simulated condition onto principal components fit on GaitNDD data (PC1 50%, PC2 38% of variance). Every simulated disease clusters together and apart from every real group; this ties back to the substrate-level double-support gap. To control for this offset, we compare the direction each simulated disease moves from simulated-healthy against the direction each real disease moves from real-healthy. 7 of 12 feature-deltas match clinical direction, which improves from 1 of 12 before calibration. This is not distinguishable from chance (binomial  $p = 0.774$ ), and the improvement is real (McNemar  $p = 0.031$ ).

This correction narrows over evaluation horizon: when recomputed on the full 1000-step baseline instead of the 500-step calibration grid, real-direction cosine similarity drops for weakness (0.828 to 0.608) and motor-noise (0.695 to 0.173) and fully inverts for feedback-noise (0.833 to -0.309), whose 80% baseline fall rate means its long-horizon average reflects pre-fall dynamics.

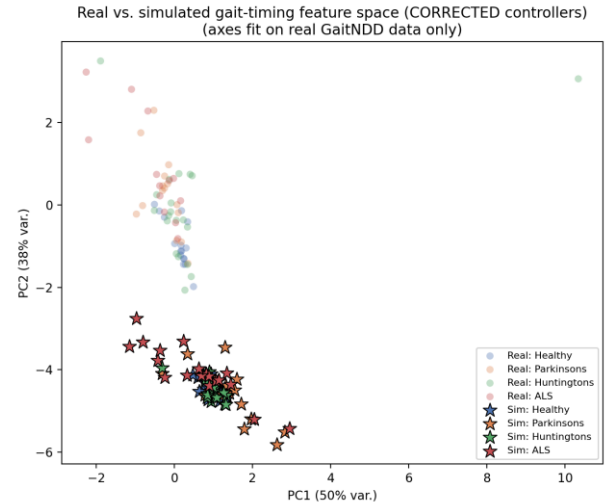


Fig. 4. Real (circles) vs. simulated (stars) gait-timing feature space; PCA axes fit on real GaitNDD data only, corrected controllers.

## IV. DISCUSSION

### Demonstrations of each construction

Kinematic imitation on clinical stroke data causes emergent clinical patterns in the model, which is shown in paretic-side muscle activation. Calibration corrects a directional bias and separates the three impairment controllers from each other on stability and push-robustness. This is because of how the shaping term is constructed. Reward-shaping additions grounded in the framework of Ng, Harada, and Russell [14] are

designed to leave a policy's optimum unchanged; they only reshape the path the optimizer takes. In other words, if the literature-reward substrate's best possible policy does not include a human-scale double-support phase, no combination of fine-tuning weight, window, or reward formulation can move that ceiling. The adversarial motion-prior fine-tune [15] fails similarly, as it changes what behavior is rewarded during training.

The comparison also raises a larger point: wherever complete kinematic data exists, like for stroke, training controllers directly on it outperforms pathological controllers built only with synthetic deficits.

The DEP-RL lineage [12], [13] established robust musculoskeletal walking; injecting impairment onto a trained policy has been done in a 2026 study which weakens plantarflexor and hip-flexor excitation on a trained gait policy but leaves validation against patient data as future work [19]. What distinguishes this study is the validation of every controller against clinical signatures, and we also map where the validation holds without assuming an injected impairment behaves like its corresponding pathology.

### Limitations

Our validation shows that the impairment-injected controllers match pathological gait distributions but doesn't completely isolate the injected mechanism as the cause of the signature. It also doesn't test predictive capability beyond the conditions used for validation. Both are directions for future work. The double-support gap's explanation (that the shared substrate's reward has no gait-timing term) was not tested by retraining from scratch, since that would have required compute resources beyond what we have. Moreover, the pelvis angle tracks its real reference poorly ( $r = -0.290$ ), which we have no clinical account for yet.

## V. CONCLUSION

This paper tested whether locomotor policy construction validated against clinical data extends from a data-rich pathology to sparser ones, and no single reward design or impairment construction won every measure. Training directly on stroke motion capture produced an emergent paretic-knee pattern that resembles stiff-knee gait. When extended to amyotrophic lateral sclerosis (ALS), Huntington's disease (HD), and Parkinson's disease (PD), whose available data is more limited, mechanism injection corrected a directional bias, but an audit falsified the idea that double-support timing is fixable at the motor-execution level. A synthetic-stroke controller isolates construction methods from diagnosis, as push-robustness collapses to the mechanism-injected range while its asymmetry direction tracks real data better than kinematic imitation. A cross-population PCA shows where the calibration helps and where not. This work shows potential for simulated stroke controllers that could better inform rehabilitation strategies and assistive-device control, as they are built on real gait data instead of hand-tuned approximations of impairment.

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